PART 1 : PHYSICS

SECTION-I

1) A particle is moving westward with a velocity $\vec{v}_1 = 5$ m/s. Its velocity changed to $\vec{v}_2 = 5$ m/s northward. The change in velocity vector $\left(\vec{\Delta V} = \vec{v}_2 - \vec{v}_1\right)$ is :

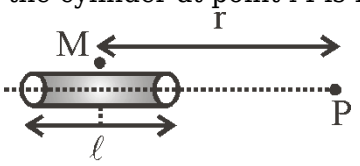
(A) $5\sqrt{2}$ m/s towards north east

(B) 5 m/s towards north west

(C) zero

(D) $5\sqrt{2}$ m/s towards north west

2) A insulating cylindrical rod of diameter d and length □ (□ >> d) has a uniform surface charge density such that the electric field just outside the curved surface of the cylinder at point M is E_0.



Find the electric field due to charge distribution at point P (r >> □).

(A) $E_0\frac{\ell d}{2r^2}$

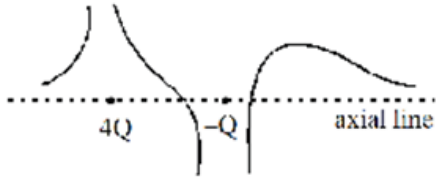
(B) $E_0\frac{\ell d}{4r^2}$

(C) $E_0\frac{\ell d}{3r^2}$

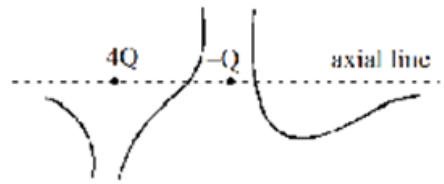
(D) $E_0\frac{2\ell d}{r^2}$

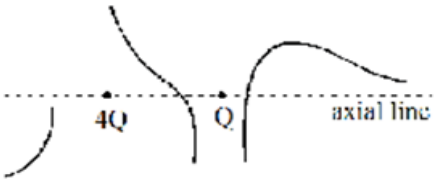
3) Graph of electric potential for two point charges 4Q and –Q along the axial line. (positive potential above the axial line and negative below the axial line).

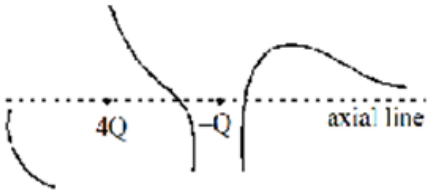
(A)



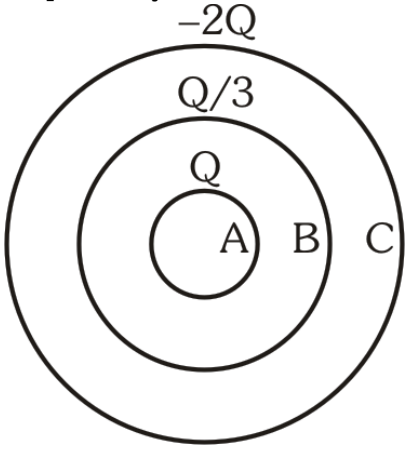
(B)



(C) 

(D) 

4) Three conducting concentric spherical shells of radius R, 2R and 3R have charges Q, $\frac{Q}{3}$ and – 2Q respectively. The intermediate shell B is now grounded. Find the charge flow into the earth.



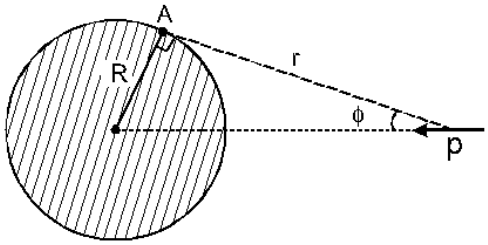
(A) $\frac{Q}{3}$

(B) $\frac{2Q}{3}$

(C) Q

(D) 0

5) A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is :



(A) $\frac{kp\cos\varphi}{r^2}$

(B) $\frac{kp\cos^2\varphi}{r^2}$

(C) zero

(D) $\frac{2kp\cos^2\varphi}{r^2}$

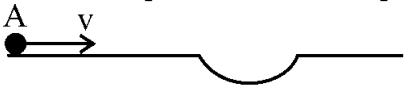
6) **Read the assertion and reason carefully to mark the correct option out of the options given below :**
Assertion: If earth suddenly stops rotating about its axis, then the value of acceleration due to gravity will become same at all the places.
Reason : The value of acceleration due to gravity is independent of rot.

(A) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(B) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(C) If assertion is true but reason is false.
(D) If the assertion and reason both are false.

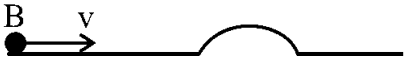
7) **Read the assertion and reason carefully to mark the correct option out of the options given below :**
Assertion: A body becomes weightless at the centre of earth.
Reason : As the distance from centre of earth decreases, acceleration due to gravity increases.

(A) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(B) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(C) If assertion is true but reason is false.
(D) If the assertion and reason both are false.

8) Two marbles A & B roll along two horizontal track with same initial speed. A moves on the track which has a dip and B moves on the track which has a bump of the same shape. Which marble wins?



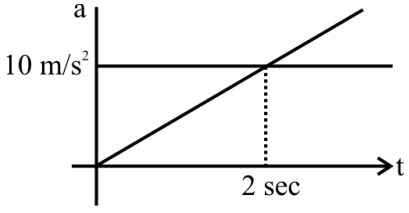
(Assume no marble leaves contact at any point.)



(A) B
(B) A
(C) Both will reach at the same time
(D) None of these

9)

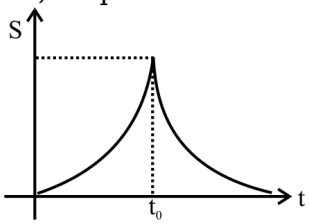
Two object starts from rest and their acceleration is shown in figure. The time when their relative velocity is again zero is :-

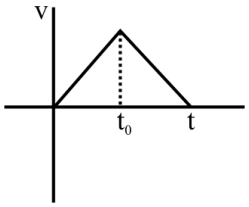


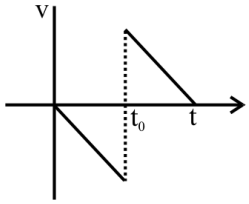
(A) 2 sec
(B) 4 sec
(C) 1 sec

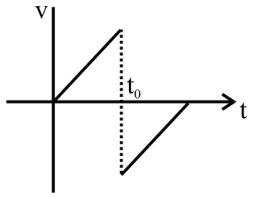
(D) 6 sec

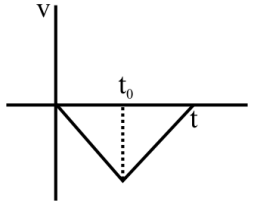
10) Displacement-time graph of an object is given. It's velocity-time graph can be:



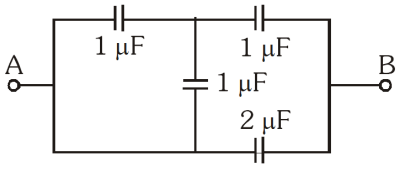
(A)


(B)


(C)


(D)


11) The equivalent capacitance between points A and B of the circuit shown will be:



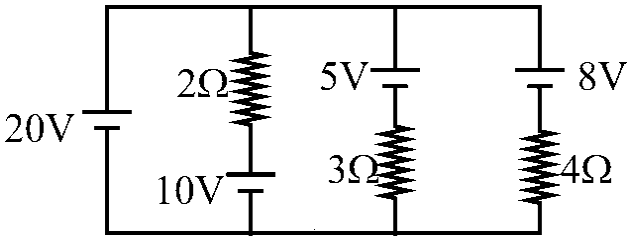
(A) $\frac{2}{3}$ μF

(B) $\frac{5}{3}$ μF

(C) $\frac{8}{3}$ μF

(D) $\frac{7}{3}$ μF

12) In the given circuit which resistor consumes maximum power :-



(A) 2Ω

(B) 3Ω

(C) 4Ω

(D) All consumes equal/over

13) The resistance of a galvanometer coil is 100 Ω. The value of current for full scale deflection in it is 1 mA. What should be the value of series resistance to be used to convert it into a voltmeter of 12 volt range

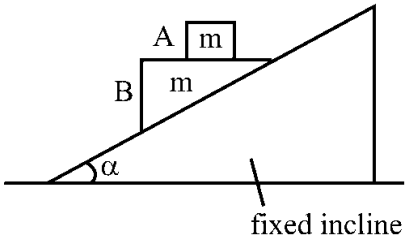
(A) 11.9 KΩ

(B) 12 KΩ

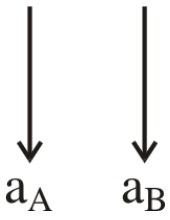
(C) 12.1 KΩ

(D) None of these

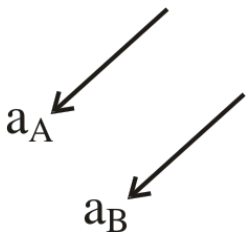
14) Two blocks A and B one on top of other, are released on a fixed incline. All surfaces are frictionless then acceleration of A & B are represented by two vectors respectively :-



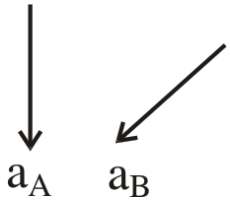
(A)



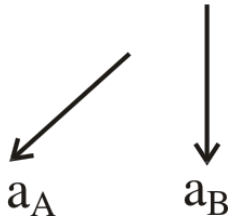
(B)



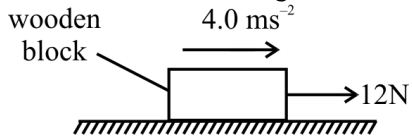
(C)



(D)



15) The given diagram shows a horizontal force of 12 N applied to a wooden block of mass 0.60 kg on a rough horizontal surface, and the block accelerating at 4.0 ms^{-2}. What is the magnitude of the frictional force acting on the block?



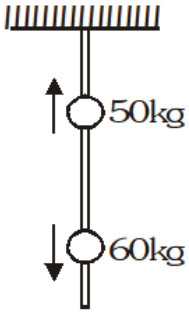
(A) 2.4 N

(B) 9.6 N

(C) 14 N

(D) 16 N

16) Two boys of masses 50 kg and 60kg are moving along a vertical massless rope, the former climbing up with an acceleration of 2ms^{-2} while later coming down with constant velocity of 2ms^{-1}.



The tension at the top of the rope connected to fixed support will be : (g = 10ms^{-2})

(A) 1.2 kN

(B) 1.1 kN

(C) 0.60 kN

(D) 1.6 kN

17) Nickel is ferromagnetic at room temperature. At a value of H field of 5×10^5 A/m, nickel saturates. The magnetic field B is 1.25 T. What is the magnetisation vector $\vec{M}$ of nickel sample?

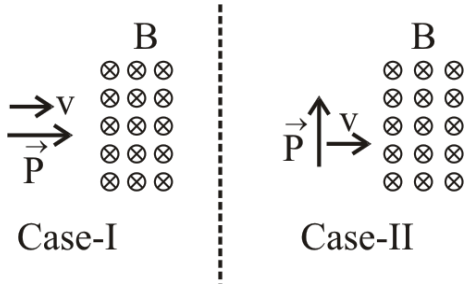
(A) 5×10^5 A/m

(B) $+3.744 \times 10^5$ A/m

(C) 9.31×10^4 A/m

(D) 15×10^5 A/m

18) An electric dipole is projected with velocity v in a uniform magnetic field directed inside the



plane of paper as shown. Choose the correct option.

(A) Dipole will experience a net force in case-I due to external magnetic field

(B) Dipole will experience a net force in case-II due to external magnetic field

(C) Dipole will experience a net torque in case-I due to external magnetic field

(D) Dipole will experience a net torque in case-II due to external magnetic field

19) Let us consider a system of unit in which units of length and force are dimensionless if mass has dimension M which of the following is incorrect ? (Hint: power = force × velocity, angular momentum = linear momentum × distance, linear momentum = mass × velocity)

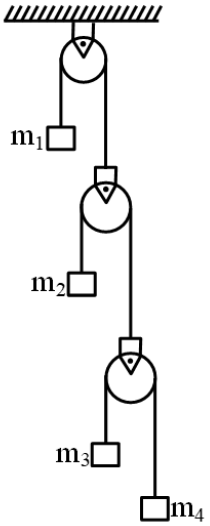
(A) Dimension of velocity is $M^{-\frac{1}{2}}$.

(B) Dimension of acceleration is M^{-1}.

(C) Dimension of power is $M^{-\frac{1}{2}}$.

(D) Angular momentum is dimensionless.

20) In the arrangement shown in figure a_1,a_2, a_3 and a_4 are the accelerations of masses m_1,m_2,m_3 and



m_4 respectively. Which of the following relation is true for this arrangement?

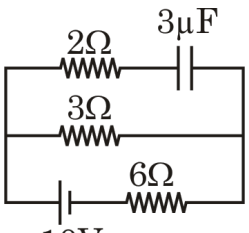
(A) $4a_1 + 2a_2 + a_3 + a_4 = 0$

(B) $a_1 + 4a_2 + 3a_3 + a_4 = 0$

(C) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

(D) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

SECTION-II

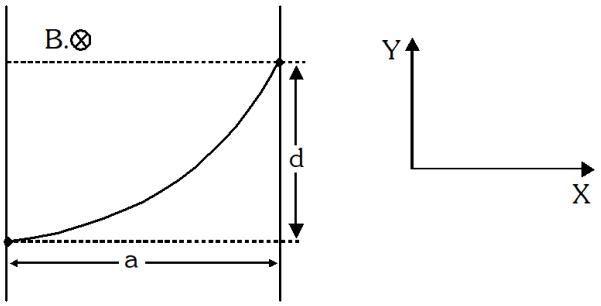


1) Find time constant of circuit shown in figure in microseconds.

2) A uniform magnetic field of magnitude 0.20 T exists in space from east to west. With what speed should a particle of mass 0.010 g and having a charge 1.0×10^{-5} C be projected from south to north so that it moves with a uniform velocity (g = 9.8 m/s^2).

3) A rod of length *l* and total charge '*q*' which is uniformly distributed is rotating with angular velocity ω about an axis passing through the centre of rod and perpendicular to rod. Find the magnitude of magnetic dipole moment (in Amp. m^2) of rod. (Take : $q = 4C$, ω = 6 rad/*s* and $l = 2m$.)

4) A particle of charge +q enters a region of uniform magnetic field B (directed into the plane of paper) as shown in the figure. Particle is deflected by distance d along Y axis after travelling a distance of along X axis. Find magnitude of linear momentum of particle in Newton-sec. (Given: a = 3m, d = 4m, Bq = 0.32C – Tesla)



5) Two parallel plate capacitors whose capacities are C and 2 C respectively, are joined in parallel. These are charged by V potential difference. If the battery is now removed and a dielectric of dielectric constant K is filled in between the plates of the capacitor C. The potential difference across each capacitor is $\frac{3V}{K+n}$. Find the value of n.

PART 2 : CHEMISTRY

SECTION-I

1) The species having pyramidal shape is :-

(A) SO_3

(B) BrF_3

(C) SiO_4^{4-}

(D) OSF_2

2) According to VBT, which of the following overlapping can result in π-type covalent bond in O_2 molecule formation, when Z-axis is internuclear axis ?
(I) 2s – 2s (II) $2p_x$ – $2p_x$ (III) 1s – 1s
(IV) $2p_y$ – $2p_y$ (V) $2p_z$ – $2p_z$

(A) I, III
(B) II, V
(C) II, IV
(D) IV, V

3) Which of the following would result in the formation of strongest p-bond if the molecular axis is x-axis ?

(A) $2p_x + 2p_x$
(B) $2p_y + 2p_y$
(C) $2p_y + 3d_{xy}$
(D) $2p_z + 4p_z$

4) Which of the following orders for electron affinity is/are correct ?
(a) S > O < Se (b) Cl > F
(c) S > O (d) O > S
(e) N > P (f) C > N

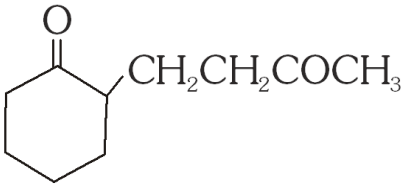
(A) a, b, c, e
(B) a, b, c, f
(C) b, c, d, e
(D) b, c, f

5) Which of the following is not correctly matched

(A) d-block element : $(n - 1)d^{1-10}\ ns^{0-2}$
(B) p-block element : $ns^2\ np^{1-6}$
(C) s-block element : ns^{1-2}
(D) Representative elements : $ns^{1-2}\ np^{1-5}$

6) Find out the incorrect statement about given molecule.
$HC \equiv C - CH = CH - CH = CH_2$

(A) All the carbons in given molecule are not in same plane
(B) IUPAC name is hexa-1,3-dien-5-yne
(C) It has 4 sp^2 hybrid carbons and 2 sp hybrid carbons
(D) All the carbons in given molecule are in same plane

7) The correct IUPAC name of  is -

(A) 4–(2–Oxobutyl) cyclohexan–1–one

(B) 2–(3–Oxobutyl) cyclohexan–1–one

(C) 1–(2–Oxocyclohexyl) cyclohexan–3–one

(D) 4–(2 Oxocyclohexyl) butan–2–one

8)

Calculate the wavelength associated with a proton moving at 1.0×10^3 ms^{-1} :-

(A) 0.032 nm

(B) 0.42 nm

(C) 2.5 nm

(D) 14.0 nm

9) The uncertainty in position of an electron is equal to its de-broglie wavelength. The minimum percentage error in its measurement of velocity under this circumstance will be approximately:-

(A) 4

(B) 8

(C) 22

(D) 18

10) Volume of O_2 required at NTP to completely burn 3g of ethane is [volume at NTP = 22.4 lit]

(A) 6.25 L

(B) 7.84 L

(C) 8.29 L

(D) 12.32 L

11) Atomic mass of element is 24 and equivalent weight is 12 then vapour density of its oxide :-

(A) 18

(B) 40

(C) 24

(D) 20

12) 1 mole of OH^- ion is obtained from 85 g of hydroxide of metal what is the equivalent weight of metal?

(A) 85 g

(B) 84 g

(C) 68 g

(D) 78 g

13)

Equivalents of MnO_4^- and $Cr_2O_7^{2-}$ per mole of the ion in acidic medium are in the ratio of :

(A) 1 : 1

(B) 1 : 5

(C) 5 : 6

(D) 6 : 1

14) $A_2 + B_2 \longrightarrow 2AB$; Rate of reaction = $k[A_2]^a[B_2]^b$

Initial $[A_2]$	Initial $[B_2]$	Rate of reaction (r) M sec^{-1}
0.2	0.2	0.4
0.1	0.4	0.4
0.2	0.4	0.8

order of reaction with respect to A_2 and B_2 are respectively :-

(A) a = 1, b = 1

(B) a = 2, b = 0

(C) a = 2, b = 1

(D) None

15) For the first order reaction, $A(g) \rightarrow B(g) + C(g) + D(s)$ taking place at constant pressure and temperature condition. Initially volume of the container containing only A was found to be V_0 and after tiome 't' it was V_t. Rate constant for the reaction is

(A) $\frac{1}{t} \ln \frac{V_0}{2V_0 - V_t}$

(B) $\frac{1}{t} \ln \frac{V_0}{V_0 - V_t}$

(C) $\frac{1}{t} \ln \frac{2V_0}{2V_0 - V_t}$

(D) $\frac{1}{t} \ln \frac{2V_t}{2V_0 - V_t}$

16) A direct current deposits 54 g of silver (Atomic mass = 108) during the electrolysis reaction. How much aluminium (Atomic mass = 27) would be deposited from aluminium chloride solution by the same amount of electricity

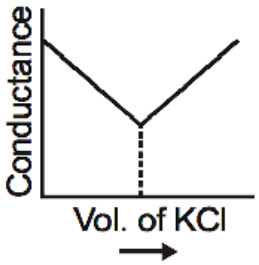
(A) 4.5 g

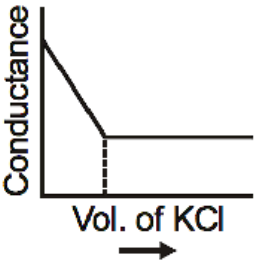
(B) 5.4 g

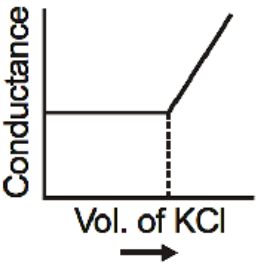
(C) 54 g

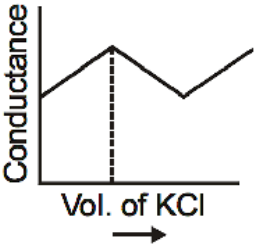
(D) 2.7 g

17) Which of following type of plot would you expect from the titration of $AgNO_3$ against KCl solution?

(A) 

(B) 

(C) 

(D) 

18) A solution of sodium sulphate in water is electrolysed using platinum electrodes. The products at cathode and anode are respectively:

(A) H_2, O_2

(B) O_2, H_2

(C) O_2, Na

(D) O_2, SO_2

19) Liquids A and B form an ideal solution in the entire composition range. At 350 K, the vapor pressures of pure A and pure B are 7×10^3 Pa and
12×10^3 Pa, respectively. The composition of the vapor in equilibrium with a solution containing 40 mole percent of A at this temperature is :

(A) $y_A = 0.37$; $y_B = 0.63$

(B) $y_A = 0.28; y_B = 0.72$

(C) $y_A = 0.76; y_B = 0.24$

(D) $y_A = 0.4; y_B = 0.6$

20) Molecules of benzoic acid (C_6H_5COOH) dimerise in benzene. 'w' g of the acid dissolved in 30 g of benzene shows a depression in freezing point equal to 2K. If the percentage association of the acid to form dimer in the solution is 80, then w is :
(Given that Kf = 5 K kg mol–1, Molar mass of benzoic acid = 122 g mol^{-1})

(A) 1.8 g

(B) 2.4 g

(C) 1.0 g

(D) 1.5 g

SECTION-II

1) 'α' particle of 3.6 Mev are fired towards nucleus ${}^{A}_{Z}X$, closest distance of approach between 'α' particle and 'X' is 1.6×10^{-14}m. Calculate atomic number of 'X'
[Given : $K = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ in S.I. units]

2) The molality of 1 litre solution with y % (w/v) of $CaCO_3$ is 2. The weight of the solvent present in the solution is 450 g then, value of y is : [At. wt. : Ca = 40 ; C = 12 ; O = 16]

3) 4 g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3. The value of x is _____ g. (Nearest integer)

4) The conductivity of pure water in a conductivity cell with electrodes of cross-sectional area 4 cm^2 and 2 cm apart is 8×10^{-7} S cm^{-1}.
The value of current that would flow through the cell under an applied potential difference of 1 volt is 1.6×10^{-X} amp. The correct value of X is:

5) The number of solution which can show negative deviation from Raoult's law is/are:
(a) Aniline + Phenol
(b) $HNO_3 + H_2O$
(c) Ethanol + n-heptane
(d) Ethanol + water
(e) Acetone + chloroform

PART 3 : MATHEMATICS

SECTION-I

1) Value of $\lim_{x\to 0} \frac{1-\cos^5 x\cos^3 2x\cos^3 3x}{x^2}$ is

(A) 6

(B) 22

(C) 11

(D) 16

2) Let $f(x) = \begin{cases} ax^2+bx+2\,; & x\geqslant 2 \\ 2ax^2+bx^3\ ; & x<2 \end{cases}$ be differentiable $\forall\, x \in \mathbb{R}$, then $\frac{|a|+|b|}{2}$ is

(A) 0.6

(B) 2.7

(C) 1.80

(D) 0.75

3) Let $f(x)$ be a continuous function in $[-1, 1]$ such that , $f(x)=\begin{cases} \frac{\ell n(px^2+qx+r)}{x^2} & ;\ -1\leq x<0 \\ 1 & ;\ x=0 \\ \frac{\sin(e^{\tan^{-1}x}-1)}{x} & ;\ 0<x\leq 1 \end{cases}$, then $(p + q + r)$ is equal to

(A) 3

(B) 2

(C) 1/3

(D) 5

4) If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then

(A) $a < 2$

(B) $2 \leq a \leq 3$

(C) $3 < a \leq 4$

(D) $a > 4$

5) Let $p, q, r \in R^+$ such that $p + q + r = 21$, then the value of $\lim_{x\to 0}\left(\frac{p^x+q^x+r^x}{3}\right)^{1/x}$ can not be equal to

(A) 7

(B) 4

(C) 8

(D) 5

6) $\lim_{x \to 0} \left(1^{\operatorname{cosec}^2 x} + 2^{\operatorname{cosec}^2 x} + 3^{\operatorname{cosec}^2 x} + ... + n^{\operatorname{cosec}^2 x}\right)^{\sin^2 x}$ is equal to

(A) n^2

(B) $\frac{n(n+1)}{2}$

(C) $n^2 + n$

(D) n

7) The term independent of x in the expansion of $\left(x^2 + x + 1 + \frac{1}{x}\right)^5$ is equal to

(A) 98

(B) 99

(C) 100

(D) 101

8) If α and β are the roots of the equation $x^2 - x + 11 = 0$, then the value of $3\alpha^3 - 3\alpha^2 + 2\beta^3 - 2\beta^2 + 11\alpha$ is

(A) 33

(B) –33

(C) 22

(D) –22

9) $S = {}^{404}C_4 - {}^{303}C_4 \cdot {}^4C_1 + {}^{202}C_4 \cdot {}^4C_2 - {}^{101}C_4 \times {}^4C_3 = (101)^k$. Find k?

(A) 1

(B) 2

(C) 3

(D) 4

10) In a study about a pandemic, data of 900 persons was collected. It was found that
190 persons had symptom of fever,
220 persons had symptom of cough,
220 persons had symptom of breathing problem,
330 persons had symptom of fever or cough or both,
350 persons had symptom of cough or breathing problem or both,
340 persons had symptom of fever or breathing problem or both,
30 persons had all three symptoms (fever, cough and breathing problem).
Number of persons having at most one symptom is

(A) 480

(B) 240

(C) 720

(D) 420

11) Coefficient of x^{2020} in $(1 + x + x^2 + x^3)^{1010} (1 - x)^{1011}$ is

(A) 0

(B) ${}^{1010}C_{505}$

(C) ${}^{-1010}C_{505}$

(D) None of these

12) For the function $f : (0, 1) \to R, f(x) = \left[2^{10}x\right] + \left\{2^6x\right\}$, the number of points of discontinuity of the function are (where [x], {x} represent G.I.F. and fractional part respectively).

(A) 900

(B) 960

(C) 1004

(D) 728

13) The set of values of p for which
$x^2 - px + \sin^{-1}(\sin 4) > 0$ for all real x is given by

(A) (–4, 4)

(B) $(-\infty, -4) \cup (4, \infty)$

(C) ϕ

(D) None

14) $f : R \to R^+$,
$f(x) = e^x$,
$g : R \to R, g(x) = 30 - 2x$ and
$f_n(x) = \underbrace{fofofo....f}_{n \text{ times}}(x)$,
then number of solution(s) of equation $f_n(g(x)) = f_{n-1}(-x)$ is

(A) 0

(B) 1

(C) 2

(D) infinite

15) $S = \dfrac{1 \times 2^2 + 2 \times 3^2 + ... + n \times (n+1)^2}{1^2 \times 2 + 2^2 \times 3 + ... + n^2 \times (n+1)} = \dfrac{an + b}{an + c}$; where a is a prime number then value of a + b + c is

(A) 9

(B) 7

(C) 6

(D) 4

16) The function f : R → R defined as $f(x) = \ell n\left(\sqrt{\sqrt{x^2+1}+x}+\sqrt{\sqrt{x^2+1}-x}\right)$ is

(A) One-one and onto both

(B) One-one but not onto

(C) Onto but not one-one

(D) Neither one-one nor onto

17) If a is even natural number, b is odd natural number and $\left[\frac{a}{b}\right]+\left[\frac{2a}{b}\right]+\left[\frac{3a}{b}\right]+...+\left[\frac{(b-1)a}{b}\right]=\frac{(a-k_1)(b-1)}{k_2}$ (where [x] represents greatest integer function of x), then $k_1 + k_2$ equals

(A) 2

(B) 3

(C) 4

(D) 5

18) The sum to infinity of the series $1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\frac{14}{3^4}+.....\infty$ is

(A) 4

(B) 6

(C) 2

(D) 3

19) The domain and range of the function $f(x)=\left[\frac{1}{\sin\{x\}}\right]$ where [x] represents greatest integer less than or equal to x and { x} = x – [x] is

(A) Domain is R, Range is Z

(B) Domain is R – Z, Range is Z

(C) Domain is R – Z, Range is N

(D) Domain is R – Z, Range is { 2, 3, 4, 5, ...}

20) Sum of twenty terms of the series 1 + (1 + 2) + (1 + 2 + 3) + ... is

(A) 1410

(B) 210

(C) 1305

(D) 1540

SECTION-II

1) If $\left(1+x+x^2\right)^{10}=\sum_{r=0}^{20} a_r x^r$ then
$a_5-\left({}^{10}C_1\right)a_4+\left({}^{10}C_2\right)a_3-\left({}^{10}C_3\right)a_2+\left({}^{10}C_4\right)a_1-\left({}^{10}C_5\right)a_0$ is equal to

2) Let $a_n \in Q$, such that $(7+4\sqrt{3})^n = a_n + b_n\sqrt{3}$, for $n \in N$. Then the value of $\left(\lim_{n\to\infty}\frac{a_n}{b_n}\right)^2$ is

3) If the number of the positive integral solutions of the equation $\left[\frac{x}{9}\right]=\left[\frac{x}{11}\right]$ is n, then $\frac{n}{3}$ is (where [.] denotes the greatest integer function)

4) Let $f:[1,\infty)-\{2\}\to B$, $f(x)=4\frac{\sqrt{x-2\sqrt{x-1}}}{(2-x)}$ is an onto function then sum of all integers in set B, is

5) If exhaustive set of values of α for which $f(x)=\alpha x^2-2\alpha x-4x+8$ is positive for exactly 3 distinct negative integral values of x is (a, b], then the value of $\frac{9a^2+b^2}{17}$ equals

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	B	A	D	B	C	C	B	B	C	C	B	A	C	B	A	A	C	D	A

SECTION-II

Q.	21	22	23	24	25
A.	12	49	4	1	2

PART 2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	D	C	B	B	D	A	B	B	B	B	D	C	C	A	A	A	C	A	B	B

SECTION-II

Q.	46	47	48	49	50
A.	20	9	1	6	3

PART 3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	B	D	B	A	C	D	D	D	D	C	C	B	C	A	A	D	B	D	C	D

SECTION-II

Q.	71	72	73	74	75
A.	0	3	8	6	1

SOLUTIONS

PART 1 : PHYSICS

1)

Conceptual

2) $E_0 = \frac{2\lambda}{2\pi\in_0 d}$

$\lambda = \pi\in_0 dE_0$

So Ep = $E_0\frac{\ell d}{4r^2}$

3)

Conceptual

4) Let a charge flows into earth

$V_2 = \frac{KQ}{2R} + \frac{K\left(\frac{Q}{3} - a\right)}{2R} + \frac{K(-2Q)}{3R} = 0$, a = 0

5)

Conceptual

6)

Conceptual

7)

Conceptual

8) For marble A, its speed increases more, so it will reach earlier.

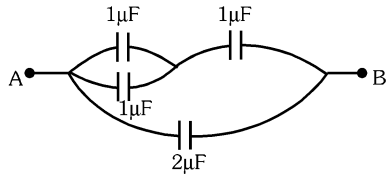
9) $\vec{v}_{rel} = 0 \Rightarrow \vec{v}_1 = \vec{v}_2$

$\Rightarrow 10t^0 = \frac{1}{2} \times t_0 \times 5t_0$

$\Rightarrow t_0 = 4$ sec

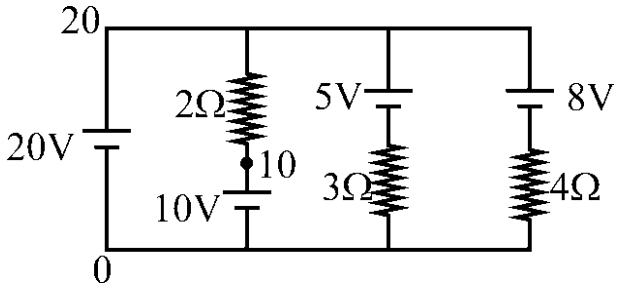
10) $v = \frac{ds}{dt}$

11)



Redraw the ckt. $C_{AB} = \frac{8}{3}$ μF

12)



R	2Ω	3Ω	4Ω
P	$\frac{10^2}{2}$	$\frac{15^2}{3}$	$\frac{12^2}{4}$

13)

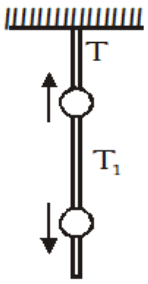
Conceptual

14) B will fall along the plane, while A will vertically downward, since force on A is only weight.

15)

$12 - f_K = 0.6 \times 4$
$f_K = 9.6$ N

16) $T_1 = 60g$



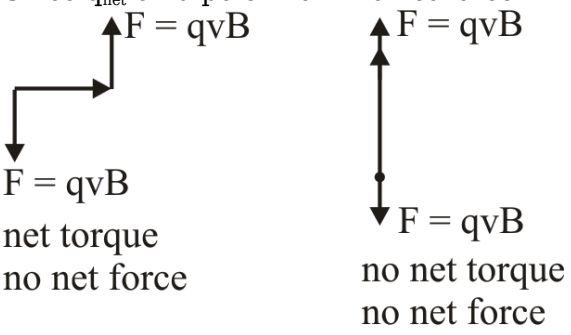
$T - T_1 - 50g$
$= 50 \times 2$
$\Rightarrow T = 110\ g + 100 = 1200$ N

17) $\vec{\mu} = \frac{\vec{B}}{\mu_0} - \vec{H}$

$= \frac{1.25}{4\pi} \times 10^7 - 5 \times 10^5 = 5 \times 10^5$ A/m.

18)

Since q_{net} on dipole = 0 ⇒ no net force



19) $[F] = [M^1L^1T^{-2}]$, $[F^0] = [M^1L^0T^{-2}]$

$\Rightarrow M^1T^{-2} = 1 \Rightarrow T = M^{\frac{1}{2}}$

(A) $[V] = [L^1T^{-1}]$,

$$[V] = \left[M^0L^0M^{-\frac{1}{2}}\right] = M^{\frac{1}{2}}$$

(B) $[a] = [LT^{-2}]$,

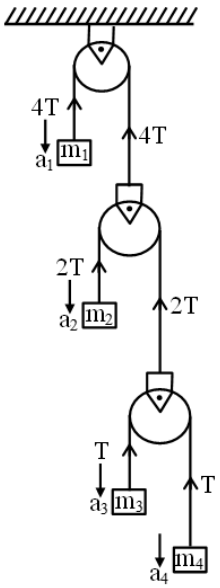
$$[a] = \left[M^0L^0\left(M^{\frac{1}{2}}\right)^{-2}\right] = M^{-1}$$

(C) $[P] = [M^1L^2T^{-3}]$,

$$[P] = \left[M^1L^0\left(M^{\frac{1}{2}}\right)^{-3}\right] = M^{-\frac{1}{2}}$$

(D) $[L] = [M^1L^2T^{-1}]$,

$$[L] = \left[M^1L^0M^{-\frac{1}{2}}\right] = M^{\frac{1}{2}}$$



20)

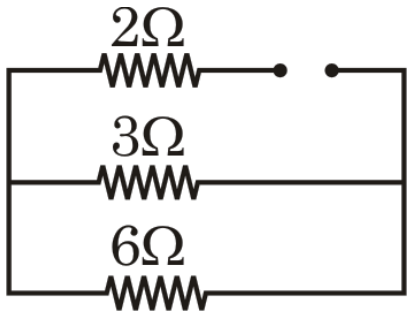
Using costraint

$\sum \vec{T}.\vec{a} = 0$

$-4Ta_1 - 2Ta_2 - Ta_3 - Ta_4 = 0$

$4a_1 + 2a_2 + a_3 + a_4 = 0$

21)



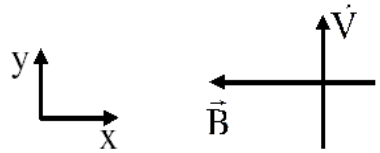
$R_{eq.} = 4\Omega$, $\tau_{eq} = 12$

22) $\vec{B} = 0.2T(\hat{i})$

m = 0.010 g

$\theta = 10^{-5}$ C

$\vec{B} = -0.2\ T\hat{i}$



$F_{net} = 0$

$qVB\hat{k} + Mg(-\hat{k}) = 0$

$Mg = qVB$

$$V = \frac{Mg}{qB} = \frac{0.01 \times 10^{-3} \times 10}{10^{-5} \times (0.2)}$$

V = 50 m/s

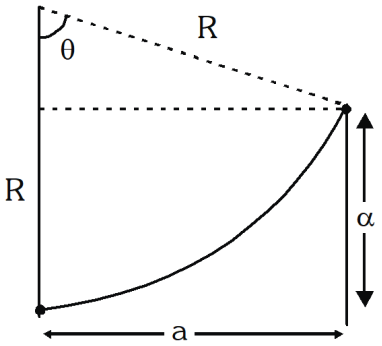
23) Current due to rotation of rod at a distance



$$x = (dq)f = \left(\frac{2q}{\ell}\right) \cdot dx \frac{\omega}{2\pi}$$

$$dM = \left(\frac{2q}{\ell}\frac{\omega}{2\pi}dx\right)\pi x^2$$

$$M = \int dM = \frac{2q}{\ell}\frac{\omega}{2} \cdot \left[\frac{x^3}{3}\right]_0^{\frac{\ell}{2}} = \frac{q\omega}{3\ell} \cdot \frac{\ell^3}{8} = \frac{q\omega\ell^2}{24}$$



24)

$a = R \sin \theta$

$d = R - R \cos \theta$

$$\left(\frac{a}{R}\right)^2 + \left(\frac{d-R}{R}\right)^2 = 1$$

$\Rightarrow a^2 + d^2 + R^2 - 2Rd = R^2$

$$\Rightarrow \frac{a^2+d^2}{2d} = R = \frac{mv}{qB}$$

$$mv = qB\left(\frac{a^2+d^2}{2d}\right)$$

$$mv = 0.32\left(\frac{3^2+4^2}{2\times 4}\right)$$

$$mv = \frac{0.32}{8}(25)$$

$mv = 1$

25) $q_1 = CV$

$q_2 = 2CV$

$q_{total} = q_1 + q_2 = 3CV$

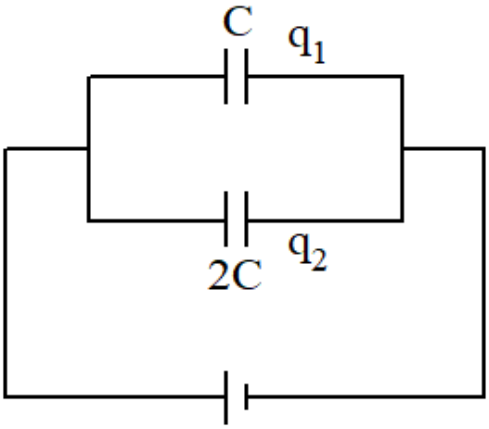
After dielectric constant 'K' is filled so

$C_{eq} = 2C + CK$

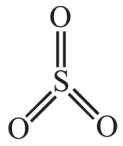
$Q = C_{eq} \times V'$

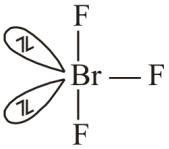
$3CV = (2C + CK)\, V'$

$$V' = \frac{3CV}{2C+CK} \Rightarrow \frac{3V}{2+K}$$

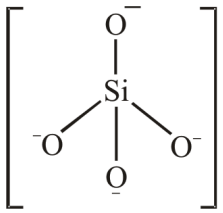


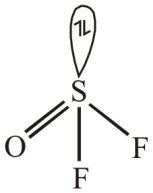
PART 2 : CHEMISTRY

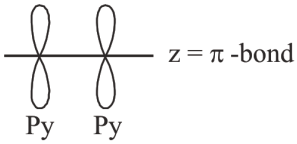




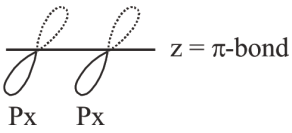
26)







27)

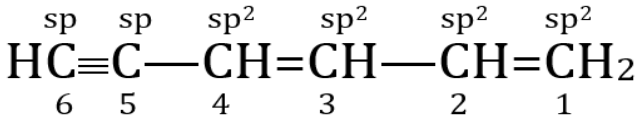


28) $2p_{\pi}$ – $2p_{\pi}$ bond in the strongest π-bond.

29) Most of the 3rd period elements have higher E.A. as compared to corresponding 2nd period element. Elements which have s^2 or p^3-configuration in outermost shell have low E.A. as compared to its next and previous element in periodic table

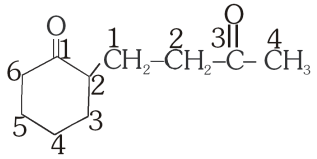
30) Representative element $ns^{1-2}np^{0-5}$

31)



hexa-1, 3-dien-5-yne

32)



2–(3–Oxobutyl) cyclohexane–1–one

33)

$$\lambda = \frac{h}{mv} = \frac{6.62 \times 10^{-34}}{1.6 \times 10^{-24} \times 1.0 \times 10^{3}} = 0.42 \text{ nm}$$

34)

$$\Delta x \cdot \Delta p = \frac{h}{4\pi} \qquad \left(\Delta x = \frac{h}{p}\right)$$

$$\frac{h}{p}\ \Delta p = \frac{h}{4\pi} \Rightarrow \frac{\Delta p}{p} = \frac{1}{4\pi}$$

$$\Rightarrow \frac{M \cdot \Delta v}{M \cdot v} = \frac{1}{4\pi}$$

$$\frac{\Delta v}{v} = 8\%$$

35) $C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

Mole of O_2 required $= \frac{7}{2} \times \frac{3}{30} = \frac{7}{20}$

Volume of O_2 required at NTP

$= \frac{7}{20} \times 22.4\,L = 7.84\,L$

36) Valency $= \frac{24}{12} = 2$

Metal oxide ⇒ MO

Molecular weight = 24 + 16 = 40

Vapour density = 40/2 = 20

37) $M(OH) \rightarrow M^+ + OH^-$

valency of metal = 1

equation wt of metal

$$= \frac{\text{weight of metal hydroxide} - \text{wt of OH}}{1}$$

$$= \frac{85 - 17}{1} = 68\ g$$

38) $MnO_4^- \rightarrow Mn^{+2}$

$Cr_2O_7^{2-} \rightarrow Cr^{+3}$

39) $r = K[A_2]^a[B_2]^b$

Put the given values of conc. and rate in equation.

40) $V_0 \propto a$

$V_t \propto a - x$

$V_t - V_0 \alpha x$
$2V_0 - V_t \alpha a - x$
$$\frac{V_0}{2V_0 - V_t} \alpha \frac{a}{a-x}$$

41) 1F = 108 g of Ag = 9g of Al
108 g of Ag $\rightarrow$ 9g of Al
54 g of Ag $\rightarrow$ x
$$x = \frac{9 \times 54}{108} = 4.5\,g$$

42) Fact

43) SRP : $Na^+ < H^+$
SOP : $SO_4^{2-} < OH^-$

44) $y_A = \dfrac{P_A}{P_{Total}} = \dfrac{P_A^o x_A}{P_A^o x_A + p_B^o x_B}$
$$= \frac{7 \times 10^3 \times 0.4}{7 \times 10^3 \times 0.4 + 12 \times 10^3 \times 0.6} = \frac{2.8}{10} = 0.28$$
$y_B = 0.72$

45) $\underset{Wg}{2(C_6H_5COOH)}$ $\underset{diner}{(C_6H_5COOH)_2}$
$\Delta_f T = i\, k_f\, m$
$$2 = 0.6 \times 5 \times \frac{w \times 1000}{122 \times 30}$$
(i = l – 0.8 +0.4 = 0.6)
w = 2.44 g

46) $$3.6 \times 1.6 \times 10^{-19} \times 10^6 = \frac{\left(9 \times 10^9\right)(z)(2)\left(1.6 \times 10^{-19}\right)^2}{1.6 \times 10^{-14}}$$
Z = 20

47) $$2 = \frac{10y}{100 \times 450} \times 1000$$
$\Rightarrow$ y = 9

Mass of NaOH = x

Moles of NaOH = $\frac{x}{40}$

Mass of Na_2CO_3 = y

Moles of Na_2CO_3 = $\frac{y}{106}$

$\frac{x}{40} = \frac{y}{106}$

x + y = 4

x = 1.1, y = 2.9

48) $x = 1.1 \approx 1$ (nearest integer)

49) $\kappa = \frac{1}{R}.\frac{\ell}{A}$

$8 \times 10^{-7} = \frac{1}{R}.\left(\frac{2}{4}\right)$

$R = \frac{10^7}{16} = 6.25 \times 10^5 \Omega$

$I = \frac{V}{R} = \frac{1}{10^7/16} = 1.6 \times 10^{-6}$amp

50)

The solution shows negative deviation from Raoult's law due to H-bond between the acetone + chloroform

Aniline + Phenol, $HNO_3 + H_2O$ shows negative deviation.

PART 3 : MATHEMATICS

51) Using expansion

$\cos^5 x \cdot \cos^3 2x \cdot \cos^3 3x =$

$$= \left(1-\frac{x^2}{2!}+....\right)^5 \cdot \left(1-\frac{(2x)^2}{2!}+....\right)^3 \cdot \left(1-\frac{9x^2}{2!}+....\right)^3$$

$$= \left(1-\frac{5x^2}{2}+....\right)\left(1-6x^2+....\right)\left(1-\frac{27x^2}{2}+....\right)$$

$$= \left(1-\left(6+\frac{5}{2}\right)x^2+....\right)\left(1-\frac{27x^2}{2}+....\right)$$

$$= 1-\frac{27x^2}{2}-6x^2-\frac{5x^2}{2}+..... = 1-22x^2+.....$$

$$\lim_{x\to 0}\frac{1-\cos^5 x\cdot\cos^3 2x\cdot\cos^3 3x}{x^2} = \lim_{x\to 0}\frac{1-(1-22x^2+....)}{x^2} = 22$$

52) For continuity

$$f(2) = \lim_{h\to 0} f(2+h) = \lim_{h\to 0} f(2-h)$$

4a + 2b + 2 = 8a + 8b

4a + 6b = 2 ⇒ 2a + 3b = 1

$$f'(x) = \begin{cases} 2ax + b & ; \quad x \geqslant 2 \\ 4ax + 3bx^2 & ; \quad x < 2 \end{cases}$$

R.H.S. = 4a + b ; LHD = 8a + 12b

8a + 12b = 4a + b

4a + 11b = 0

$$b = -\frac{4a}{11}$$

$$2a - \frac{12a}{11} = 1 \Rightarrow a = \frac{11}{10};\ b = -\frac{4}{10}$$

53) r = 1 and $\lim_{x\to 0^-} \ell n\left(1 + qx + px^2\right)^{\frac{1}{x^2}} = 1$

□n(e^p) = 1 ⇒ p = 1, q = 0

54) Roots are real and less than specific number 3

$f(x) = x^2 - 2ax + a^2 + a - 3$

(i) D ≥ 0

⇒ $4a^2 - 4(a^2 + a - 3) \geq 0$

⇒ $-4a + 12 \geq 0$

⇒ $4a - 12 \leq 0$

⇒ $a \leq 3$ ..(1)

(ii) f(3) > 0

⇒ $9 - 6a + a^2 + a - 3 > 0$

⇒ $a^2 - 5a + 6 > 0$

⇒ $(a-2)(a-3) > 0$

⇒ $\boxed{a \in (-\infty, 2) \cup (3, \infty)}$ ...(2)

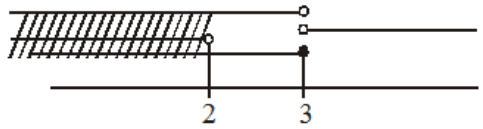
(iii) $\frac{-b}{2a} < 3$

⇒ $\frac{2a}{2} < 3$

⇒ $\boxed{a < 3}$ ...(3)

Now,

Taking Intersection of set of values of 'a' obtained from (1), (2) and (3)



⇒ $a \in (-\infty, 2)$

⇒ $\boxed{a < 2}$

55) Let $L = \lim_{x\to 0}\left(\frac{p^x + q^x + r^x}{3}\right)^{\frac{1}{x}}$ → (1^∞ form)

$= e^{\lim_{x\to 0}\frac{1}{x}\left(\frac{p^x+q^x+r^x}{3}-1\right)}$

$= e^{\lim_{x\to 0}\frac{1}{3}\left[\left(\frac{p^x-1}{x}\right)+\left(\frac{q^x-1}{x}\right)+\left(\frac{r^x-1}{x}\right)\right]}$

$= e^{\frac{1}{3}[\ell np+\ell nq+\ell nr]}$

$= e^{\frac{1}{3}\ell n(pqr)} = (pqr)^{1/3}$

Now apply A.M. ≥ G.M.

$\frac{p+q+r}{3} \geqslant (pqr)^{1/3} \Rightarrow \frac{21}{3} \geqslant L \Rightarrow L \leqslant 7$

∴ L cannot be 8

56) $\lim_{x\to 0} n\left(\left(\frac{1}{n}\right)^{\cos ec^2 x} + \left(\frac{2}{n}\right)^{\cos ec^2 x} + ... + 1\right)^{\sin^2 x} = n$

57) Coeff of x^0 in

$\frac{(x^3+x^2+x+1)^5}{x^5} = \frac{(x^2+1)^5(x+1)^5}{x^5}$

∴ Coeff of x^5 in $(x^2 + 1)^5 (x + 1)^5$

= Coeff of x^5 in $(1 + 5x^2 + 10x^4 + ...x^{10})(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5) = 1 + 50 + 50 = 101$

58) $\alpha^2 - \alpha + 11 = 0 \Rightarrow 3\alpha^3 - 3\alpha^2 + 33\alpha = 0$ ...(i)

$\beta^2 - \beta + 11 = 0$ $2\beta^3 - 2\beta^2 + 22\beta = 0$ ...(ii)

(i) + (ii)

$\Rightarrow 3\alpha^3 - 3\alpha^2 + 2\beta^3 - 2\beta^2 + 33\alpha + 22\beta = 0$

$\Rightarrow 3\alpha^3 - 3\alpha^2 + 2\beta^3 - 2\beta^2 + 11\alpha = -22(\alpha + \beta) = -22$

59) $S = ({}^{101}C_1)^4 = (101)^4$

60)

n(U) = 900

Let A ≡ Fever, B ≡ Cough

C ≡ Breathing problem

∴ n(A) = 190, n(B) = 220, n(C) = 220

$n(A \cup B) = 330$, $n(B \cup C) = 350$,

$n(A \cup C) = 340$, $n(A \cap B \cap C) = 30$

Now $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow 330 = 190 + 220 - n(A \cap B)$

$\Rightarrow n(A \cap B) = 80$

Similarly,

$350 = 220 + 220 - n(B \cap C)$

$\Rightarrow n(B \cap C) = 90$

and $340 = 190 + 220 - n(A \cap C)$

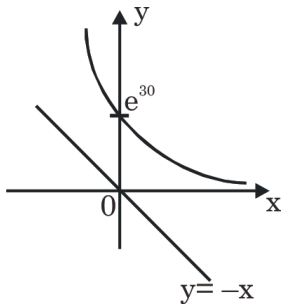
$\Rightarrow n(A \cap C) = 70$

 n(A∪B∪C) = (190 + 220 + 220) – (80 + 90 + 70) + 30
= 660 – 240 = 420
⇒ Number of person without any symptom
= n (∪) – n(A ∪ B ∪ C)
= 900 – 420 = 480
Now, number of person suffering from exactly one symptom
= (n(A) + n(B) + n(C)) – 2(n(A ∩ B) + n(B ∩ C) + n(C ∩ A)) + 3n(A ∩ B ∩ C)
= (190 + 220 + 220) – 2(80 + 90 + 70) + 3(30)
= 630 – 480 + 90 = 240
 Number of person suffering from atmost one symptom
= 480 + 240 = 720

61) $((1 + x + x^2 + x^3)^{1010} (1 - x)^{1011}$
$= (1 - x^4)^{1010} (1 - x)$
Required coefficient : $-{}^{1010}C_{505}$

62) $f : (0, 1) \to R, f(x) = \left[2^{10}x\right] + \left\{2^6x\right\}$,
$[2^{10}x]$ is discontinuous at x = 1, 2, 3,.... 1023
$\{2^6x\}$ is discontinuous at x = 1, 2, 3,.... 63
but $f(x) = \left[2^{10}x\right] + \left\{2^6x\right\}$ is continuous at x = 1, 2, 3, ... 63
⇒ Points of discontinuity = 1023 – 63 = 960

63) $x^2 - px + \pi - 4 > 0 \ \forall \ x \in R$
$D = p^2 - 4(\pi - 4) < 0 \Rightarrow P^2 + 16 - 4\pi < 0$
Cannot be negative for any value of $p \in R$ set
values of p = ϕ.



64) $\underbrace{f o f o f o ... f}_{n \text{ times}}(g(x)) = \underbrace{f o f o f .... o f}_{n-1 \text{ times}}(-x)$
⇒ f(g(x)) = –x
⇒ $e^{30-2x} = -x$ ⇒ No solution

65) $S = \dfrac{\Sigma r(r+1)^2}{\Sigma r^2 (r+1)} = \dfrac{\Sigma r^3 + 2\Sigma r^2 + \Sigma r}{\Sigma r^3 + \Sigma r^2}$

$$S = \frac{\left[\frac{n(n+1)}{2}\right]^2 + \left[\frac{n(n+1)(2n+1)}{3}\right] + \left[\frac{n(n+1)}{2}\right]}{\left[\frac{n(n+1)}{2}\right]^2 + \left[\frac{n(n+1)(2n+1)}{6}\right]}$$

$$S = \frac{\frac{n(n+1)}{2} + \frac{2(2n+1)}{3} + 1}{\frac{n(n+1)}{2} + \frac{2n+1}{3}}$$

$$S = \frac{3\left(n^2+n\right) + 8n + 4 + 6}{3\left(n^2+n\right) + 4n + 2} = \frac{3n^2 + 11n + 10}{3n^2 + 7n + 2}$$

$$S = \frac{(3n+5)(n+2)}{(3n+1)(n+2)} = \frac{3n+5}{3n+1}$$

66)
$$= \frac{1}{2}\ell n\left(\sqrt{\sqrt{x^2+1}+x} + \sqrt{\sqrt{x^2+1}-x}\right)^2$$

$$= \frac{1}{2}\ell n\left\{\sqrt{x^2+1} + x + \sqrt{x^2+1} - x + 2\sqrt{\left(\sqrt{x^2+1}\right)^2 - x^2}\right\}$$

$$= \frac{1}{2}\ell n\left\{2\sqrt{x^2+1} + 2\right\}$$

An even function hence neither one-one nor onto.

67) Given series

$$\left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \left[\frac{3a}{b}\right] + ... + \left[\frac{(b-1)a}{b}\right]$$

have (b – 1) terms

First term + last term $= \left[\frac{a}{b}\right] + \left[\frac{ab}{b} - \frac{a}{b}\right]$

$= \left[\frac{a}{b}\right] + a + \left[-\frac{a}{b}\right]$

$= a - 1$

Similarly

second term + second last term

$= \left[\frac{2a}{b}\right] + \left[(b-2)\frac{a}{b}\right]$

$= \left[\frac{2a}{b}\right] + \left[a - \frac{2a}{b}\right]$

$= \left[\frac{2a}{b}\right] + a + \left[-\frac{2a}{b}\right]$

$= a - 1$

So required sum $= \frac{(b-1)}{2}(a-1)$

$\Rightarrow k_1 + k_2 = 3$

68)
$$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + ...$$

$$\frac{S}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + ...$$

$$\frac{2}{3}S = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + ...$$

$$= \frac{4}{3} + \frac{4}{9}\left(\frac{1}{1-\frac{1}{3}}\right)$$

$$S = \frac{3}{2}\left(\frac{4}{3} + \frac{4}{9} \times \frac{3}{2}\right)$$
S = 3

69) Domain
$\sin\{x\} \neq 0$
$\therefore \{x\} \neq 0$
$\Rightarrow x \neq \text{integer}$
$D_f \in R - Z$
Range $\{x\} \in (0, 1)$
$\sin\{x\} \in (0, \sin 1)$
$\frac{1}{\sin\{x\}} \in (\text{cosec}\, 1, \infty)$
$\left[\frac{1}{\sin\{x\}}\right] \in \{1, 2, 3, 4, ..\}$

70) $S = \sum_{n=1}^{20} T_n$
$= \sum_{n=1}^{20} \frac{n(n+1)}{2}$
$= \sum_{n=1}^{20} \frac{n^2}{2} + \sum_{n=1}^{20} \frac{n}{2}$
$= \frac{1}{12}(20 \times 21 \times 41) + \frac{1}{4}(20 \times 21)$
=1540.

71) Given sum = coefficient of x^5 in expansion of $(1-x)^{10}\left(1 + x + x^2\right)^{10}$ which is $\left(1 - x^3\right)^{10}$.
So required sum = 0 (as there is no any term containing x^5)

72) $(7 + 4\sqrt{3})^n = a_n + b_n\sqrt{3} \Rightarrow (7 - 4\sqrt{3})^n = a_n - b_n\sqrt{3}$
$\Rightarrow a_n = \frac{1}{2}[(7 + 4\sqrt{3})^n + (7 - 4\sqrt{3})^n]$ and $b_n = \frac{1}{2\sqrt{3}}[(7 + 4\sqrt{3})^n - (7 - 4\sqrt{3})^n]$
$\Rightarrow \lim_{n\to\infty} \frac{a_n}{b_n} = \sqrt{3} \Rightarrow \left(\lim_{n\to\infty} \frac{a_n}{b_n}\right)^2 = 3$

73) $\left[\frac{x}{9}\right] = \left[\frac{x}{11}\right] = l$
$\Rightarrow l \leqslant \frac{x}{9} < l + 1,\ l \leqslant \frac{x}{11} < l + 1,$ Solution is possible only when $11l < 9l + 9$
$\Rightarrow 0 \leqslant l < \frac{9}{2}$
Total number of solution
$\sum_{l=0}^{4} (9 - 2l) - 1 = 24$

74) $f(x) = 4\dfrac{\sqrt{\left(1-\sqrt{x-1}\right)^2}}{2-x} = 4\dfrac{\left|1-\sqrt{x-1}\right|}{2-x}$

$$= \begin{cases} 4.\dfrac{1-\sqrt{x-1}}{2-x} = \dfrac{4}{1+\sqrt{x-1}} & 1 \le x < 2 \\ 4.\dfrac{\sqrt{x-1}-1}{2-x} = \dfrac{-4}{1+\sqrt{x-1}} & x > 2 \end{cases}$$

if $x \in [1, 2)$, then $y \in (2, 4]$
if $x \in [2, \infty)$, then $y \in (-2, 0)$
sum = 3 + 4 – 1 = 6 Ans.

75)

$f(x) = (\alpha x - 4)(x - 2) \Rightarrow f(x) = 0$ has roots $\dfrac{4}{\alpha}, 2$
α must be negative and

$$\frac{4}{\alpha} \in [-4,-3) \Rightarrow \frac{\alpha}{4} \in \left(\frac{-1}{3}, \frac{-1}{4}\right] \Rightarrow \alpha \in \left(\frac{-4}{3}, -1\right]$$

$$\Rightarrow a = \frac{-4}{3}, b = -1 \Rightarrow 9a^2 = 16, b^2 = 1 \Rightarrow 9a^2 + b^2 = 17$$